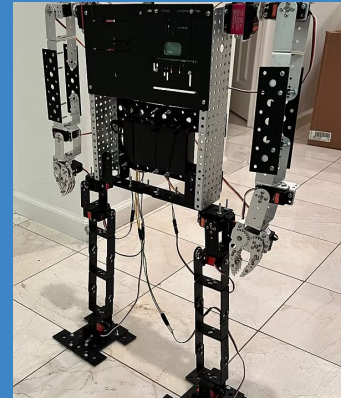


Gurdwara Gur Nanak Prakash Tracy Robotics Team

Week 1

About Me

- Lead Field Technician for Quarry Lane High School Robotics Team
- Part of the Design & Electrical Division
- Conducted research on tactile training for quadrupeds with Professor Cao from Case Western Reserve University
- Alameda County Science Fair
 - Humanoid Robot to Fight Fires → 2nd Place
 - Tree Planting Robot to Combat Climate Change → 1st Place



What is FTC and How to Get to Worlds?

- FTC = FIRST Tech Challenge → build, program, drive 18 inch by 18 inch robots
- Teams compete in alliances of 2 vs 2 with autonomous and remote control period
- Norcal → Teams go to league meets regionals and the best robots advance to Norcal Championship → Best Teams are awarded with chance to compete at

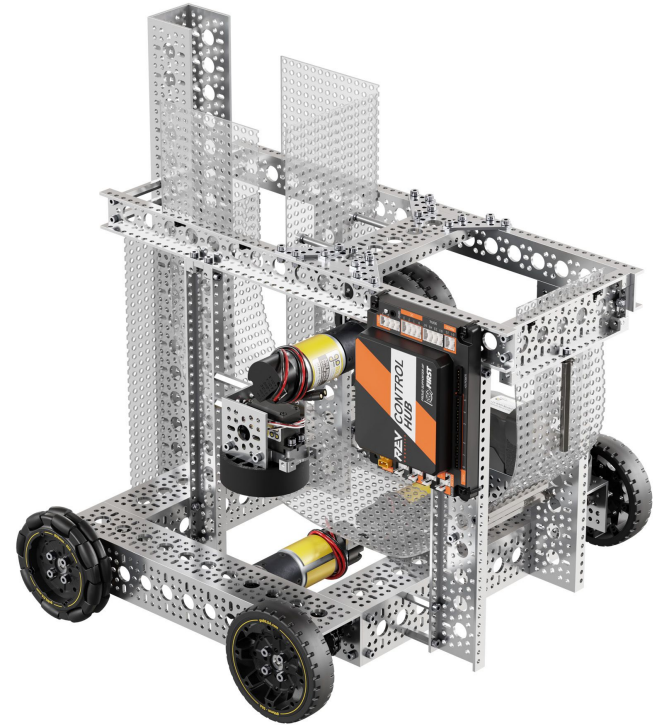
2025-2026 World Championship in Houston



	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.
Registration Opens												
Pre-Season												
Kickoff												
Build & Practice Season												
Qualifying Season												
State/Regional Championships												
FIRST Championship												

The Tentative Plan

- Understand Game Rules
- Determine Game Strategy → How can we maximize points?
- Start working on preliminary design based on strategy
- Begin Fabrication and Assembly
- Look to compete in Early January



REV Hub Not Included

Game for 2025-2026



Mock Game for 2025-2026



Team Roles Hardware

Hardware

- Design → Work on designing the robot using CAD software based on the strategy discussed → Show preliminary CAD
- Electrical → Responsible for wiring the robot and calibrating all the electrical components
- Assembly → Responsible for assembling the robot(almost everyone will do this)

Team Roles Software

Software

- Drivetrain → Responsible for coding robot motion
- Intake → Responsible for coding the robot intake
- Shooter → Responsible for coding the shooting of the game piece
- Strategy → Responsible for collaborating with alliance partners at competitions to decide on how the specific game should be played

Next Steps

- Begin Registration of the FTC team and apply for any available grant
- Work to get a field setup for us to begin practicing on, once our robot is ready
- More information will be sent out this week about our next meetings
- For students:
 - Complete this form if you have not: <https://forms.gle/Bw5BdLajfYCLNteD7>
 - Review these resources from the FIRST Tech Challenge:
 - FTC Documentation: <https://ftc-docs.firstinspires.org/en/latest/>
 - FTC Decode Game Manual: <https://ftc-resources.firstinspires.org/ftc/game/manual>